

From a Featureless Global Observable to Mode-Resolved Candidate Inverse Features

Theory, validation, and reconnaissance results for the hydroelastic forward model

Technical writeup

This note documents the reconnaissance stage of this project: the search for observables of the wet hydroelastic response suitable as candidates for the inverse (parameter-recovery) problem. Findings are marked in green, open or scope-limited points in red, and background definitions in blue, following the convention used in the companion sensor-placement writeup for this project.

1 Motivation

The ultimate goal of this project is to infer a floating ice floe’s hydroelastic properties: the nondimensional flexural-stiffness parameter β , mass/draft parameter γ , and radius R , from sparse buoy IMU time series (currently a four-sensor array). The intended route is: fit a state-space model to the measured signals with stochastic subspace identification (SSI), extract observable features from that fit, and use a sensitivity Jacobian $\partial \mathbf{f} / \partial (\beta, \gamma, R)$ to test whether those features actually constrain the three unknown parameters (a local identifiability question, to be answered by an SVD of the Jacobian in later work). That whole programme depends, at its very first step, on having a forward-model observable $\mathbf{f}(\beta, \gamma, R)$ that is (a) computable from the already-validated hydroelastic forward model, (b) sensitive to the parameters, and (c) not so featureless or so broad that a finite-difference derivative would be dominated by numerical noise.

The first candidate tried was the simplest possible one: the magnitude of the total spatial response, $\|w(\omega)\|$, swept over a broad frequency range at the nominal floe parameters. This is a reasonable default choice, and it is exactly analogous to the trace of an observability Gramian in this project’s companion sensor-placement study: both are a single aggregate *sum* over many underlying degrees of freedom. That analogy turns out to be the whole story of this document: a sum, by construction, cannot penalise or highlight any single weak term, and the total-norm reconnaissance found no useful resonance structure across the investigated range at all. Section 2.2 makes this precise. The resolution, decomposing w onto the ice floe’s own dry natural modes so that each mode’s contribution can be examined *individually* rather than swallowed into one number, is the technical core of everything that follows, and it is what actually uncovered usable structure.

2 Background theory

2.1 The hydroelastic forward problem

Physical setup. The floe is modelled as a thin, free-edge circular elastic plate of radius R , floating on water of depth H , forced by a normally-incident, time-harmonic plane wave of radian frequency ω . The unknown is the complex plate deflection $\eta(r, \theta; \omega)$ (polar coordinates centred on the floe), governed by the coupled plate-fluid boundary value problem: the thin-plate (Kirchhoff) equation for the plate,

$$D \nabla^4 \eta - \rho_i h \omega^2 \eta = p(r, \theta) \quad (r < R), \quad (1)$$

where D is the flexural rigidity, $\rho_i h$ the floe’s mass per unit area, and p the hydrodynamic pressure at the plate’s underside; linearised water-wave (Laplace) equations in the fluid, with the standard free-surface and seabed conditions away from the floe; and continuity of pressure and vertical velocity across the plate-water interface, plus free-edge conditions (zero bending moment, zero Kirchhoff effective shear) at $r = R$.

Non-dimensionalisation. Following Meylan & Squire (1996) and Montiel’s later extensions, the problem is made non-dimensional using the water depth H , giving three governing parameters in

addition to the geometric (R, ν) :

$$\alpha = \frac{\omega^2 H}{g}, \quad \beta = \frac{D}{\rho_w g H^4}, \quad \gamma = \frac{\rho_i h}{\rho_w H}, \quad (2)$$

where ρ_w is the water density and g gravitational acceleration. α is the non-dimensional forcing frequency; β is a non-dimensional flexural stiffness; γ is a non-dimensional mass (draft-related) parameter. **These three parameters, β, γ, R , are exactly the physical unknowns this whole project is trying to recover.** Everything in this document is computed at a sweep of α (equivalently ω) values, at a single fixed trial (β, γ, R) : the nominal floe parameters $\beta_0 = 4.6985 \times 10^{-5}$, $\gamma_0 = 1.4548 \times 10^{-3}$, $R_0 = 0.3830$, $\nu = 0.3$.

Matched eigenfunction expansion (EMM). The forward model (this project’s `ForwardModel/deflection.m` and its supporting functions) solves Equation (1) together with the fluid problem by the matched eigenfunction method: the velocity potential is expanded, in each spatial region (open water outside the floe, disc-covered water beneath it), in a complete set of vertical eigenfunctions built from the roots of a dispersion relation $D(\kappa; \alpha, \beta, \gamma) = 0$: one travelling root, a set of complex, damped roots, and infinitely many purely evanescent roots, truncated in practice to M terms. The plate deflection is expanded radially, in each vertical eigenfunction, using Bessel functions appropriate to that root’s character (modified Bessel $I_n(\kappa_m r)$ for the purely evanescent roots, with the travelling and complex-damped roots entering through their own corresponding radial forms), so the disc-covered region is not treated as evanescent-only, and angularly in $\cos(n\theta)$ (for a wave incident along $\theta = 0$, only the cosine family is populated by symmetry: there is no roll/antisymmetric component). The expansion coefficients are fixed by enforcing continuity and the plate’s edge conditions at $r = R$, a linear system built from inner-product (orthogonality) matrices between the two regions’ eigenfunctions, truncated at orders M (evanescent modes), P (edge/ Gegenbauer terms), and N (angular orders). The result is a deflection field

$$\eta(r, \theta; \alpha, \beta, \gamma, R, \nu) \quad (\text{complex-valued, truncated at } M, P, N). \quad (3)$$

This machinery, and its truncation-convergence behaviour, is extensively unit-tested and numerically validated elsewhere in the project (`ForwardModel/Unit-Tests/`) and is treated here as a known-good black box that produces η given a frequency and trial parameters.

2.2 Why a global response norm can hide real structure

The most naive scalar observable one could sweep over frequency is the spatial L^2 norm of the field itself,

$$\|\eta(\omega)\|^2 = \int_0^{2\pi} \int_0^R |\eta(r, \theta; \omega)|^2 r dr d\theta. \quad (4)$$

This is exactly the object swept (at a set of discrete spatial evaluation points, as a proxy for the integral) by this project’s initial reconnaissance script. Section 2.4 below constructs an orthonormal modal basis $\{w_{n,j}\}$ for the deflection field, with modal coefficients $A_{n,j} = \langle \eta, w_{n,j} \rangle$. Parseval’s identity for an orthonormal basis gives an exact decomposition of Equation (4):

$$\|\eta(\omega)\|^2 = \sum_{n,j} |A_{n,j}(\omega)|^2. \quad (5)$$

Equation (5) is exact for the continuous L^2 norm. The original reconnaissance script instead swept a discrete spatial proxy for that integral, so the sum-hides-weak-terms argument below is approximate, not an exact identity, when applied to that script’s actual output.

The sum-hides-weak-terms argument. Equation (5) says the global norm is literally a *sum* of modal energies. This is mathematically the same situation as the trace of an observability Gramian in this project’s companion sensor-placement study, which is also a sum ($\text{tr}(W_o) = \sum_i \lambda_i$) and was noted there to be unable to penalise a single poorly-observed direction, since a sum does not

care which terms are small. Here the parallel is exact: if one modal term (in practice the rigid heave term $A_{0,0}$, by far the largest-amplitude contribution at low frequency) dominates the sum in Equation (5), then $\|\eta(\omega)\|$ tracks $|A_{0,0}(\omega)|$ almost exclusively, and a genuine, frequency-localised maximum or near-zero in a smaller term such as $A_{2,0}$ or $A_{0,1}$ can be numerically present in the sum and yet be completely invisible to the eye (or to a peak-finding algorithm) in the aggregate curve. A sum can only reveal a feature that is large *relative to the sum itself*, not one that is large relative to its own typical scale.

What the global-norm reconnaissance actually found. Swept broadly over α at the nominal parameters, $\|\eta(\omega)\|$ showed no clear, frequency-localised resonance peaks: a smooth, essentially monotonic decay dominated by the explicit $1/\alpha$ kinematic prefactor in the deflection field’s own definition. Diagnostics built into that reconnaissance script, removing the known $1/\alpha$ scaling and separately checking the field’s *phase* (which a pole crossing must swing through, while a fixed kinematic prefactor cannot), were used specifically to guard against concluding “no resonance” when the real issue was an amplitude scaling artefact obscuring one. (*Figure placeholder: insert the $\|\eta(\omega)\|$ reconnaissance plot from `nominalFRFREconnaissance.m` here if available.*) This established, before any further work, that the aggregate norm alone was not a suitable observable for extracting localised spectral features of this kind, and motivated the modal decomposition below.

2.3 The dry plate eigenmodes

The free-edge biharmonic eigenproblem. Following Montiel and the classical plate-vibration literature (e.g. Leissa, 1969), the *dry*, in-vacuo natural modes of a free-edge circular plate solve

$$\nabla^4 W - \frac{\lambda^4}{R^4} W = 0, \quad W(r, \theta) = R_{n,j}(r) \begin{Bmatrix} \cos n\theta \\ \sin n\theta \end{Bmatrix}, \quad (6)$$

with $\lambda = kR$ a non-dimensional eigenvalue and n the angular (nodal-diameter) order. The radial part is built from ordinary and modified Bessel functions, $R_{n,j}(r) = J_n(\lambda r/R) + C I_n(\lambda r/R)$, with λ and C fixed by imposing the two free-edge conditions (zero radial bending moment M_r , zero Kirchhoff effective shear V_r) at $r = R$; this gives a 2×2 homogeneous system whose non-trivial solutions require $\det B_n(\lambda) = 0$, a transcendental eigenvalue equation solved numerically, root by root, for each n . **Rigid-body modes** ($n = 0$, heave, $R(r) = 1$; $n = 1$, tilt, $R(r) = r$) are not roots of this determinant at all (the free-edge equations degenerate identically at $\lambda = 0$) and are constructed directly instead. Each mode is numerically normalised so that $w_{n,j} = R_{n,j}(r) \cos(n\theta)$ (or the sin partner) has unit L^2 norm over the disc.

The dry basis is validated to essentially machine precision. Free-edge boundary residuals and cross-mode orthogonality ($\langle w_{n,j}, w_{n,j'} \rangle = 0$ for $j \neq j'$ at fixed n) were confirmed numerically at the level of floating-point round-off (`circularPlateModeTester.m`). Independently, the computed dry-mode eigenvalues $K_{n,j}$ were benchmarked against values obtained from Montiel’s free-edge eigenvalue formulation (Montiel, 2012, PhD thesis, University of Otago, Ch. 6, Eq. (6.2)) and agreed to within approximately 0.12% maximum relative error. This matters beyond simple code-correctness: it means the basis used for the projection in Section 2.4 is not merely self-consistent, it reproduces an external, independently derived reference to high precision.

Exact benchmark source to be confirmed. The 0.12% figure above should be cross-checked against the specific reference values actually used inside `circularPlateModeTester.m` before it is quoted in the final report, to confirm whether those values were taken from a printed table in Montiel’s thesis or recomputed independently from his characteristic equation.

2.4 Orthogonal projection onto the dry basis

The projection. Given the (validated, orthonormal) dry-mode basis $\{w_{n,j}\}$ of Section 2.3, the

wet EMM field $\eta(r, \theta; \omega)$ is decomposed by ordinary orthogonal projection,

$$A_{n,j}(\omega) = \langle \eta(\cdot; \omega), w_{n,j} \rangle = \int_0^{2\pi} \int_0^R \eta(r, \theta; \omega) R_{n,j}(r) \cos(n\theta) r dr d\theta. \quad (7)$$

Because the basis is orthonormal, projecting the theta-integral first via the angular Fourier orthogonality of $\cos(n\theta)$ collapses Equation (7) into one theta integral per distinct n (not per mode) followed by a cheap radial quadrature per j (`projectEMMOntoPlateModes.m`); an implementation detail, but one that makes a full multi-mode, multi-frequency sweep computationally tractable.

Best-approximation property. Orthogonal projection onto a finite subspace is, by a standard Hilbert-space fact, the *best* approximation of η achievable using only that subspace: reconstructing $\eta_{\text{recon}} = \sum_{(n,j) \in S} A_{n,j} w_{n,j}$ from any finite mode set S minimises the L^2 reconstruction error over all possible choices of coefficients for that mode set. A direct and useful consequence is **monotonicity under nesting**: if $S_1 \subset S_2$, the reconstruction error using S_2 can never exceed that using S_1 , since S_2 's best approximation has strictly more freedom available to it. This is not an empirical hope; it is a guaranteed property of projecting onto an orthonormal basis, and it is exactly what should be (and was) observed as more dry modes were retained.

The dry basis is an efficient representation of the real wet field. Projecting the actual nominal-parameter EMM field onto Montiel's eight dominant dry modes ($A_{n,0}$, $n = 0-4$; $A_{n,1}$, $n = 0-2$) and reconstructing from those eight coefficients alone reproduces the true field to within approximately 2% relative L^2 error. Extending the retained mode set to 24 modes reduces this to

$$\boxed{\text{relative reconstruction error} \approx 0.36\%}. \quad (8)$$

This is a substantial result in its own right, independent of anything found later in the frequency sweep: it demonstrates that a handful of dry, in-vacuo plate modes (computed with *no* fluid coupling at all) already span almost all of the genuinely wet, hydroelastic response. The dry basis is therefore not just a mathematically convenient decomposition; it is an efficient one. (*Figure placeholder: insert a reconstruction-error-vs-mode-count plot here if available, from ForwardModel/ValidationExperiments/singleFrequencyModalProjectionTest.m.*)

Three distinct quantities, not to be conflated. From here on this document works with three related but distinct objects: the *dry* natural frequencies of Section 2.3 (an in-vacuo plate property, no fluid at all); the *wet modal-response features* extracted below (Section 3), which are properties of the forced EMM field at the nominal trial parameters; and, later, *poles recovered by SSI* from a fitted state-space model of a simulated or measured time series. These may turn out to be numerically close to one another, but nothing in this document establishes that, and the sections below are written so as not to assume it.

2.5 Resonance-like maxima versus modal near-zeros: two different spectral features

Two qualitatively different kinds of frequency-localised feature appear in the results below, and it is worth being precise about why they are different, since both are useful but for different reasons.

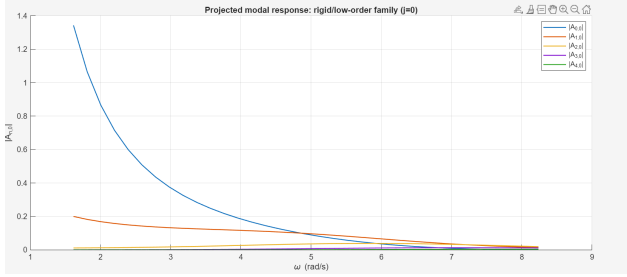
Resonance-like maxima. A local maximum of $|A_{n,j}(\omega)|$ is the signature classically associated with a resonance: for many physically damped or radiating systems, a pole of the modal coefficient close to the real ω -axis (viewed as a function of complex frequency) produces such a bump as ω sweeps past it. That mechanism is descriptive here, not established: whether these particular maxima correspond to an actual pole of the wet, coupled system remains to be tested, for instance against free-decay or SSI-extracted dynamics, and no such correspondence is assumed in what follows. These features are consistently called *modal-response maxima* or *resonance-like maxima*, never confirmed poles.

Modal-coefficient near-zeros and phase-reversal features. A local *minimum* of $|A_{n,j}(\omega)|$ that goes essentially to zero, accompanied by a phase change of approximately 180° immediately either side of it, is a structurally different object from a maximum: it indicates that the smooth complex curve $A_{n,j}(\omega)$ in the complex plane passes through (or very close to) the origin, i.e. $\text{Re } A_{n,j}$ and $\text{Im } A_{n,j}$ both nearly vanish at the same frequency. A complex-valued coefficient does not have a sign in the ordinary sense, so this document calls the feature a *near-zero / phase-reversal feature* rather than a “sign change”. It is a reproducible feature of the chosen incident-wave-to-modal-response map, computed at fixed incident-wave conditions, and may therefore provide a useful parameter-sensitive observable; whether its location is independent of the forcing configuration is not tested here. Because $|A| \rightarrow 0$ makes the phase numerically ill-conditioned (the argument of a near-zero complex number is dominated by round-off), the phase is only diagnostic *away from* the immediate minimum, in the roughly 180° swing either side of it, not as a meaningful value exactly at the minimum itself.

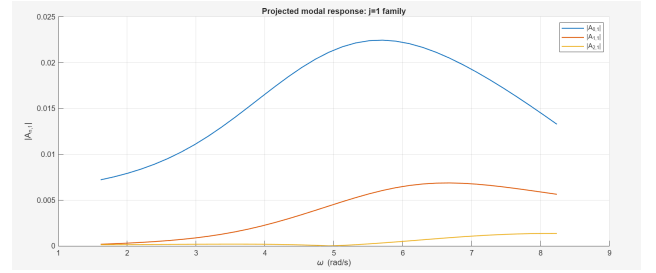
3 Reconnaissance results

3.1 The coarse 35-point sweep exposes mode-specific structure

With the dry basis and projection validated, a 35-point sweep in physical ω (kept inside the truncation-validated range, $\alpha \lesssim 13$) was run at the nominal parameters, solving the EMM field once per frequency and projecting onto Montiel’s eight dominant modes at each (`modalRA0Reconnaissance.m`). The complex $A_{n,j}(\omega)$ curves, not merely their magnitudes, were retained throughout, since Section 2.5’s near-zero/phase-reversal features depend on phase as much as amplitude.



(a) Rigid/low-order family, $|A_{n,0}|$



(b) $j = 1$ family, $|A_{n,1}|$

Figure 1: Coarse 35-point modal-response magnitude sweep. The $j = 0$ family is dominated by the large, monotonically decaying rigid heave term $A_{0,0}$ (note the individual small-amplitude terms $A_{3,0}, A_{4,0}$ are visually flattened by this: exactly the sum-hides-weak-terms effect of Section 2.2, here acting even within a single combined-scale plot, not only in the fully aggregated norm). The $j = 1$ family shows clear broad maxima in $A_{0,1}$ and $A_{1,1}$, absent from the aggregate norm.

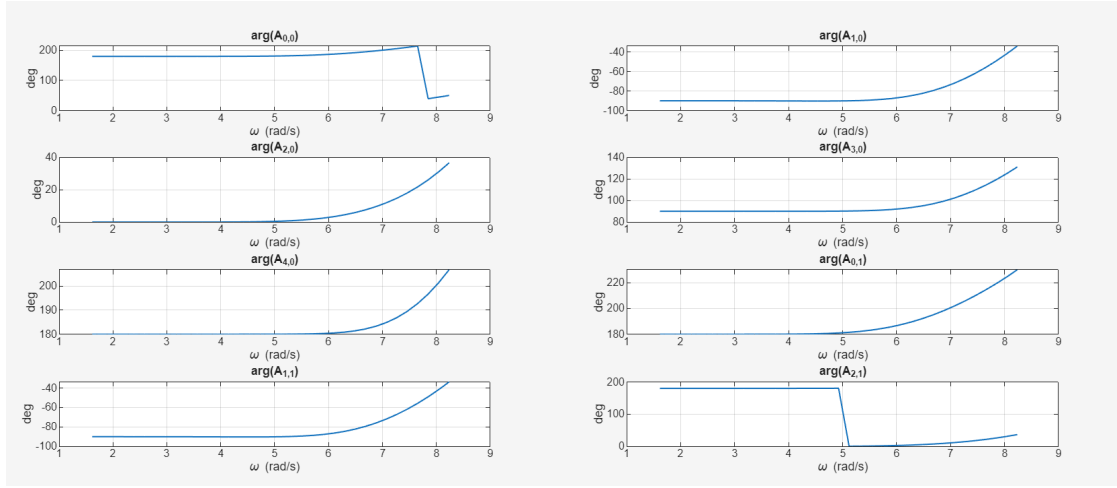


Figure 2: Coarse-sweep phase, $\arg A_{n,j}(\omega)$, unwrapped, one panel per mode. Most modes show smooth, gradual phase rotation, consistent with the sweep being numerically well-behaved (no evidence of `dispersionRoots.m` branch-switching in this range). Two panels are qualitatively different: $\arg A_{2,1}$ shows an abrupt $\sim 180^\circ$ swing near $\omega \approx 5$, and $\arg A_{0,0}$ shows a similar abrupt swing near $\omega \approx 7.8$: both are the near-zero/phase-reversal signature of Section 2.5, not numerical artefacts, and both were followed up by local refinement below.

The modal decomposition recovered exactly the kind of structure the global norm could not show. $A_{2,0}$ grows from low frequency to a broad maximum before decaying; $A_{0,1}$ and $A_{1,1}$ each show a clear, broad maximum at a mode-specific frequency; $A_{2,1}$ falls to a magnitude minimum with an accompanying phase reversal, a genuine modal near-zero rather than a resonance; and the same near-zero/phase-reversal signature was subsequently found, unexpectedly, in the dominant heave term $A_{0,0}$ near the top of the validated frequency range. None of this is visible in $\|\eta(\omega)\|$.

3.2 Local refinement and the current candidate feature set

The 35-point grid is too coarse to trust apparent extremum locations directly (a broad maximum in particular can shift by a full grid step under refinement, which would make a finite-difference Jacobian built on it artificially quantised). Each candidate feature was therefore re-swept on a much finer local ω -grid, targeted only at its own neighbourhood (`modalFeatureRefinement.m`); not a repeat of the global sweep. For the $A_{0,0}$ candidate near-zero, which sits close to the truncation orders' validated ceiling, the same local window was additionally re-solved at a bumped truncation order $(M, P, N) = (50, 10, 10) \rightarrow (70, 15, 15)$ on the *identical* frequency grid (so that any shift in the result reflects genuine truncation-order dependence, not grid quantisation), as a check against the feature being a numerical artefact rather than a real one.

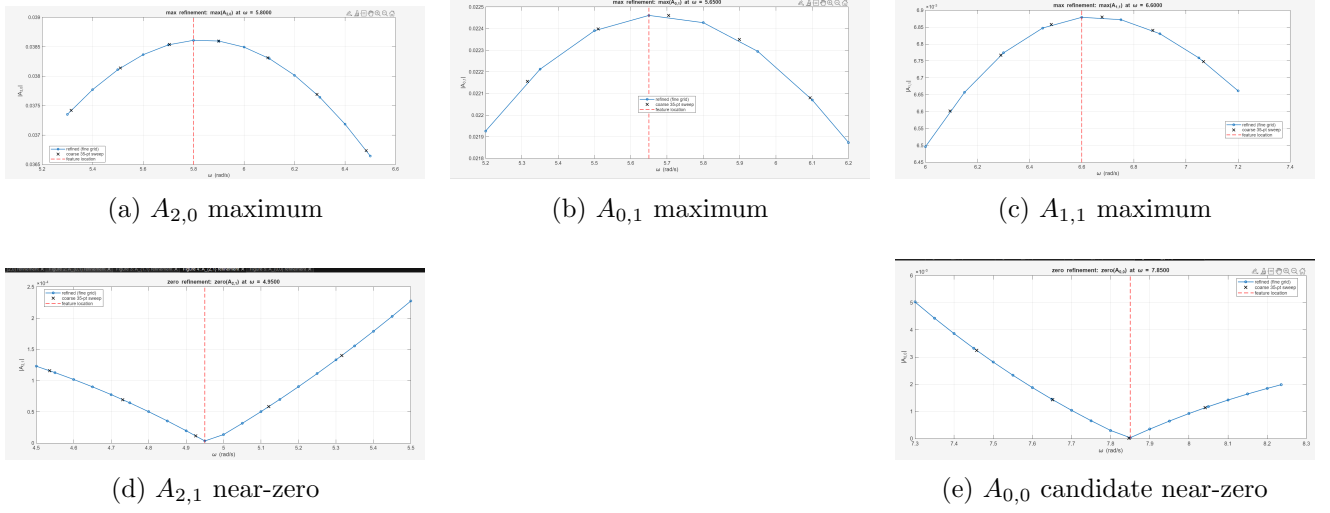


Figure 3: Locally refined modal-response curves (blue, fine grid) with the original coarse 35-point samples overlaid (black crosses) and the extracted feature location marked (red dashed line). All five features fall comfortably inside their refinement window, away from the window edge.

Mode	Feature	Location	Status
$A_{2,0}$	$\omega_{\max}$	≈ 5.80 rad/s	interior maximum on the refined window
$A_{0,1}$	$\omega_{\max}$	≈ 5.65 rad/s	interior maximum on the refined window
$A_{1,1}$	$\omega_{\max}$	≈ 6.60 rad/s	interior maximum on the refined window
$A_{2,1}$	ω_0	≈ 4.95 rad/s	near-zero + $\sim 180^\circ$ phase reversal; $ A _{\min} \approx 3.5 \times 10^{-6}$
$A_{0,0}$	ω_0	≈ 7.85 rad/s	same grid location under tested truncation increase $(50, 10, 10) \rightarrow (70, 15, 15)$

Terminology, deliberately conservative. These are called *modal-response maxima* (or *resonance-like maxima*) and *modal-coefficient near-zero/phase-reversal features* throughout, not “natural frequencies” and not “SSI poles”. A dry-mode natural frequency (Section 2.3), a wet modal-response extremum (this section), and a pole later recovered from measured data by SSI are three conceptually distinct quantities that may turn out to be numerically close, but nothing established so far demonstrates that; conflating them would be a claim this document does not make.

The broad-maximum caveat. $A_{0,1}$ and $A_{1,1}$ ’s maxima in particular are broad and comparatively flat (Figure 3b,c). A naive discrete $\arg \max$ feature extractor is therefore a risk for the planned finite-difference Jacobian: a small parameter perturbation could leave the same grid point as the maximum (derivative artificially zero), then cross a threshold and jump by a full grid step. The planned fix, already agreed but not yet implemented, is sub-grid parabolic interpolation: fitting $y(\omega) = a\omega^2 + b\omega + c$ to $y = |A|^2$ at the three points bracketing each discrete extremum and taking the vertex $\omega_\star = -b/2a$ as a continuous estimate, evaluated at a *fixed* triple of frequencies across parameter trials rather than re-searched each time, so that the observable varies smoothly with (β, γ, R) instead of being locked to a grid.

3.3 Qualitative consistency with the published literature

Montiel et al. (2013, *J. Fluid Mech.* 723, 629–652), figure 3, report resonance-type peaks in the experimentally and numerically obtained $A_{2,0}$ and $A_{0,1}$ modal RAOs, and their $A_{2,1}$ curve shows a pronounced minimum/phase-change-type feature. The curves obtained here show the same *qualitative* pattern: maxima in $A_{2,0}$ and $A_{0,1}$, and a near-zero/phase-reversal feature in $A_{2,1}$.

This is a qualitative consistency check, not quantitative validation. The floe parameters, water depth, and normalisation convention used here differ from Montiel’s published case, so the specific frequency values are not expected to (and do not need to) match numerically. What this check establishes is that the general qualitative pattern of which modes carry resonance-like structure and which carry near-zero-type structure is consistent with an independently published,

physically-equivalent hydroelastic model: a mild but genuine corroboration that these are not artefacts specific to this implementation.

4 Discussion

4.1 Central finding

Mode-resolved observables reveal distinct, frequency-localised candidate inverse features that are hidden by the global response norm.

This is not an implementation detail; it is the scientific result of this stage of the project. Section 2.2 provides the mathematical mechanism by which this masking occurs (Parseval’s identity, Equation (5), together with the dominance of the rigid heave term), Sections 2.3–2.4 establish that the dry modal basis used to expose this is independently validated and an efficient representation of the real wet field, and Section 3 shows the structure this decomposition actually recovers: three resonance-like maxima and two near-zero/phase-reversal features, one of which ($A_{2,1}$) is a strongly resolved numerical feature of the EMM response, and one of which ($A_{0,0}$) is a new observation not anticipated at the start of this reconnaissance.

4.2 What has, and has not, been established

Established: the dry, free-edge circular-plate eigenbasis is correct to machine precision internally and matches values computed from Montiel’s published free-edge formulation to $\sim 0.12\%$ (exact source to be re-confirmed, Section 2.3); this dry basis reconstructs the real, wet, hydroelastically-coupled EMM field to within $\sim 2\%$ error using only eight modes and $\sim 0.36\%$ using 24; the resulting modal decomposition of a 35-point frequency sweep reveals clear, mode-specific frequency-localised structure that is completely absent from the aggregate response norm; five candidate features (three maxima, two near-zero/phase-reversal features) have been extracted and locally refined, all falling safely inside their refinement window; the $A_{2,1}$ near-zero reaches $|A|_{\min} \approx 3.5 \times 10^{-6}$ together with an $\sim 180^\circ$ phase reversal; the $A_{0,0}$ candidate near-zero remains at the same grid location, $\omega = 7.85$ rad/s, under the tested truncation increase $(M, P, N) = (50, 10, 10) \rightarrow (70, 15, 15)$ on the same frequency grid, ruling out the simplest form of numerical artefact; and the general pattern of which modes carry maxima versus near-zero-type features is qualitatively consistent with Montiel et al. (2013).

Not established, and explicitly out of scope for this stage:

1. *Sub-grid feature precision.* The three modal-maximum locations are currently discrete-grid estimates; the parabolic sub-grid refinement described in Section 3.2 has been designed but not yet implemented or run.
2. *Parameter sensitivity.* Nothing in this document yet tests how any of the five candidate features move under perturbation of (β, γ, R) . That is precisely the subject of the next stage (`modalFeatureVector.m` and the finite-difference sensitivity Jacobian), not addressed here.
3. *Identifiability.* Whether these five observables, even if individually well-behaved, jointly constrain all three unknown parameters (as opposed to being insensitive to one of them, or degenerate with each other) is an SVD/conditioning question for the Jacobian stage, not something a forward-model reconnaissance can answer on its own.
4. *Connection to SSI.* These are forward-model, EMM-side observables computed from the known deflection field. Whether, or how well, a state-space model fitted to a simulated or real noisy time series by SSI can actually recover features at or near these same frequencies is untested and is explicitly a later validation step (this project’s SSI-bridge work), not assumed here.
5. *Generality beyond the nominal parameters.* All results in this document are at the single nominal trial (β_0, γ_0, R_0) . Whether the same qualitative picture (which modes show resonance-like maxima, which show near-zero features, roughly where) persists across the parameter space the

4.3 Summary and next steps

This stage of the project replaced a global response norm that is an unsuitable observable for extracting localised spectral features (a featureless, sum-dominated aggregate norm) with a validated, mode-resolved alternative, and used it to identify five concrete candidate features with real, checkable evidence behind each of them, not merely plausible-looking bumps on a coarse curve. The dry-mode basis underlying this decomposition is independently validated against both internal consistency checks and Montiel’s published free-edge formulation, and was shown to be an efficient (not merely convenient) representation of the true wet response. The investigation deliberately stops at this point, having extracted and refined a concrete feature set rather than pushing further into denser sweeps or additional modes, because the natural next question, whether these features actually constrain the unknown floe parameters and how well-conditioned that constraint is, requires a different kind of experiment, not a finer frequency sweep. That next experiment is `modalFeatureVector.m`: fixed three-frequency anchors with sub-grid quadratic interpolation for each feature location, followed by central finite differences at $\epsilon = 10^{-3}, 10^{-4}, 10^{-5}$ to check step-size robustness, suitable non-dimensional scaling of the resulting Jacobian J_s , and an SVD of J_s to assess local identifiability of (β, γ, R) .

References

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